The role of motion in deep video analysis

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Computer vision researcher
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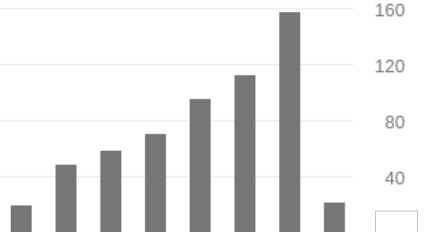
Computer Vision Video understanding Motion analysis Future anticipation

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TITLE	CITED BY	YEAR
Video acceleration magnification Y Zhang, SL Pintea, JC Van Gemert Proceedings of the IEEE Conference on Computer Vision and Pattern …	112	2017
Deja Vu: Motion Prediction in Static Images SL Pintea, JC van Gemert, AWM Smeulders Proceedings of the European Conference on Computer Vision (ECCV) 2014, 172-187	83	2014
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- PhD in Computer Vision at UvA with Arnold Smeulders in 2017
- R&D Engineer at Blippar/Layar on augmented reality
- Postdoc at TUDelft on tremor analysis and geophysical imaging
- Researcher at TUDelft on efficiency in deep learning
- Senior postdoc at LUMC on surgery video analysis

- **Supervision**: 15 MSc students, 2 PhDs (co-promotor)
- **Teaching** (BKO diploma):
  - Course manager: BSc Image Processing
  - Co-lecturer: MSc Computer Vision by Deep Learning

Why am I talking about videos?

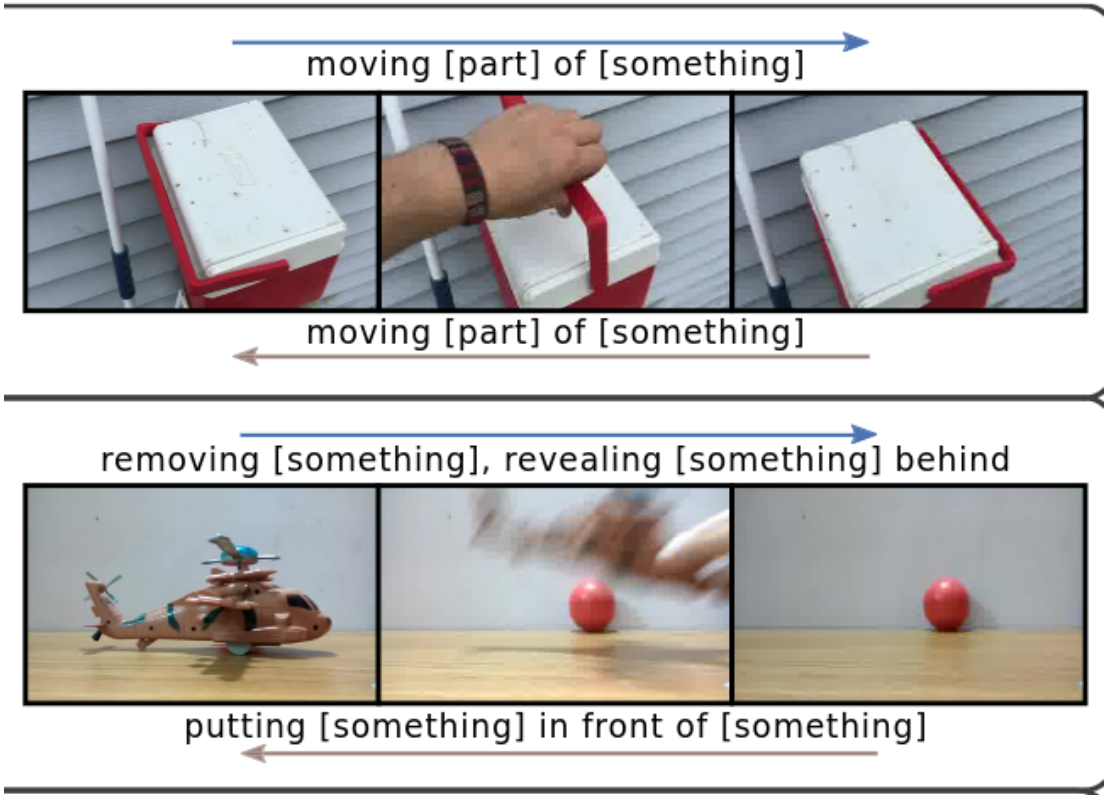
- To better allocate resources (nurses, operation rooms, cleaning teams, etc):
  - How far along the surgery is? (e.g. 45%)
  - What is the current surgery phase? (e.g. administering anesthetic, opening up the patient, etc.)

Video from the public MVOR dataset

Why motion?

Why motion?

- Differentiate visually similar actions:

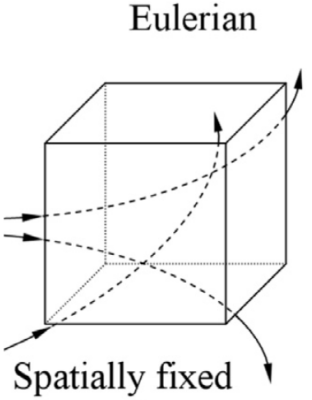


[*Retro-Actions: Learning 'Close' by Time-Reversing 'Open' Videos*. W. Prince and D Damen, ICCVw, 2019]

Eulerian versus Lagrangian motion

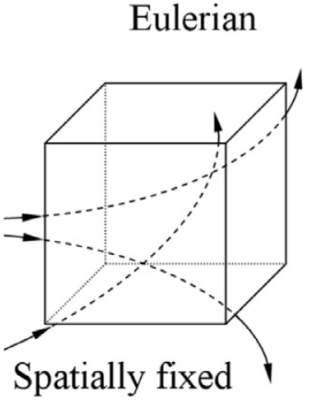
Eulerian versus Lagrangian motion

- **Eulerian motion:** how the video changes over time at a fixed point in space

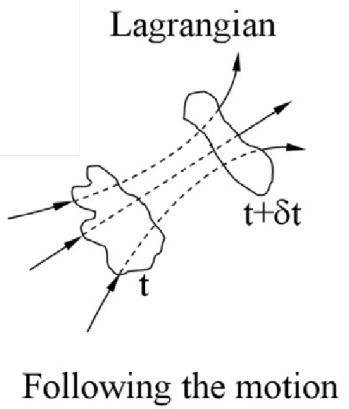


Eulerian versus Lagrangian motion

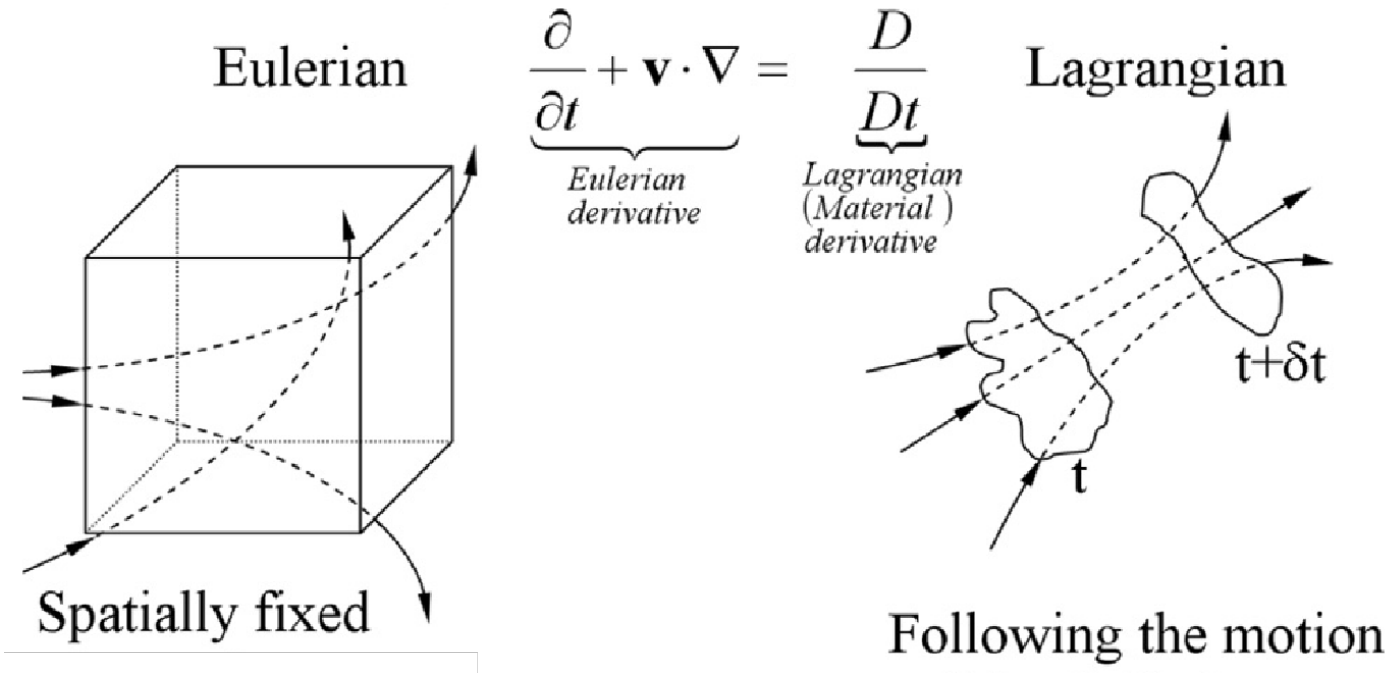
- **Eulerian motion:** how the video changes over time at a fixed point in space



- **Lagrangian motion:** tracking an individual "video voxel" as it moves through space and time



Eulerian versus Lagrangian motion

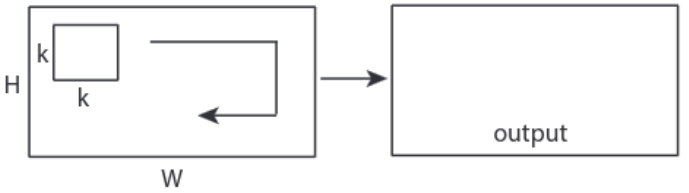


[*Smoothed particle hydrodynamics method for fluid flows, towards industrial applications: Motivations, current state, and challenges*. Shadloo, Oger, Le Touze, 2016]

Eulerian motion in video analysis

Eulerian motion in video analysis

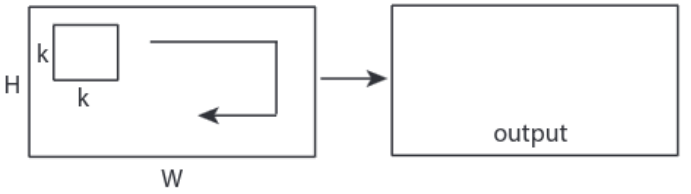
- 3*D* convolutions: process video volumes



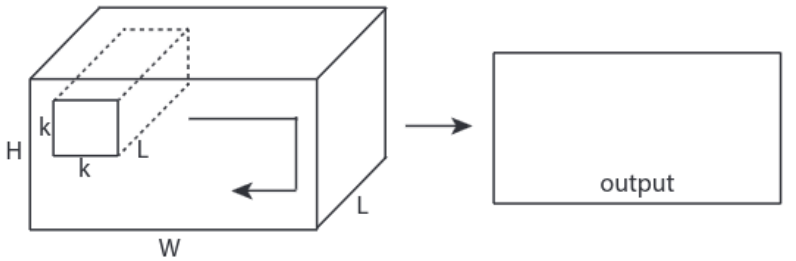
(a) 2D convolution

Eulerian motion in video analysis

- 3*D* convolutions: process video volumes



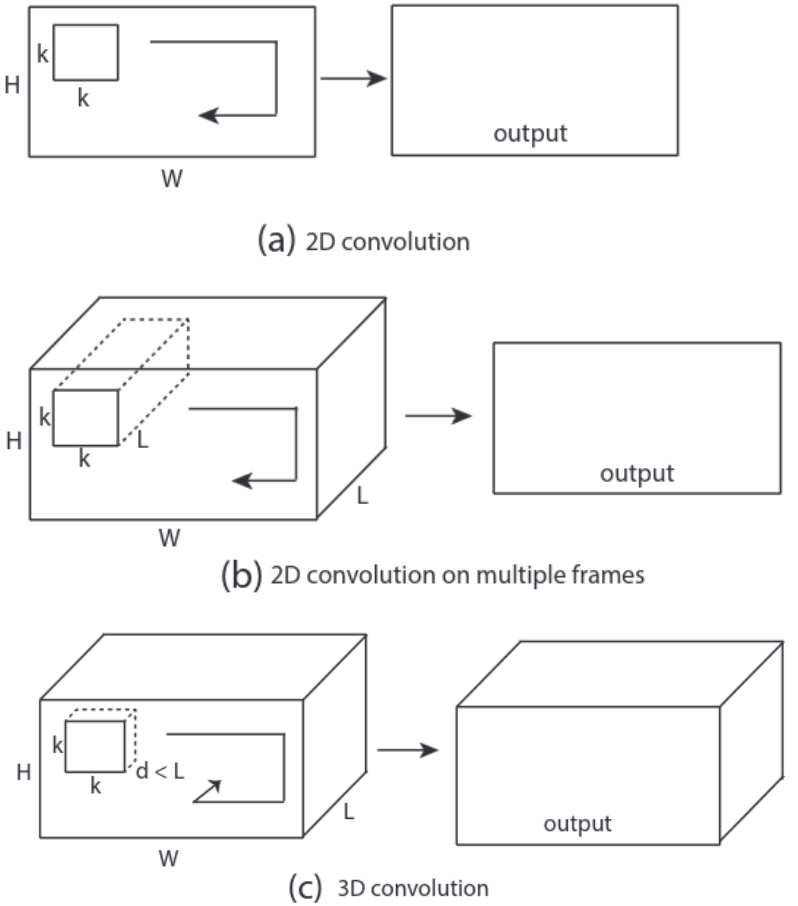
(a) 2D convolution



(b) 2D convolution on multiple frames

Eulerian motion in video analysis

- 3*D* convolutions: process video volumes



[Tran, Du, et al. *"Learning spatiotemporal features with 3d convolutional networks."* ICCV 2015.]

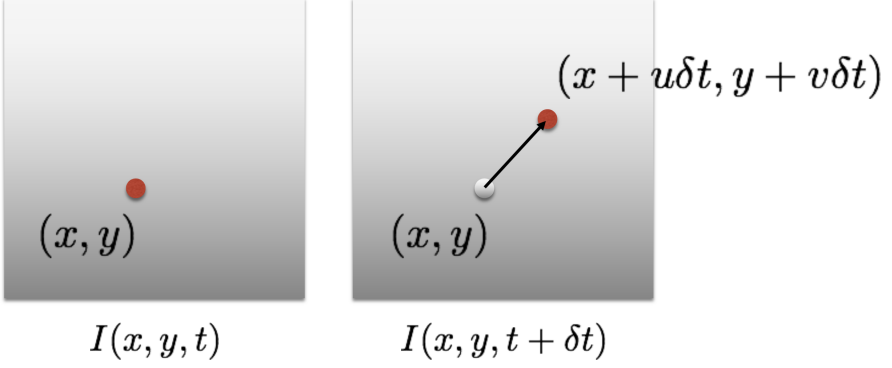
Lagrangian motion in video analysis

- Typically use *optical flow* as input to the video analysis model

Lagrangian motion in video analysis

- Typically use *optical flow* as input to the video analysis model

- Optical flow gives the motion vectors (u, v) at every frame pixel (x, y) over time, t



From CMU Computer Vision Lecture 24, Optical flow

Lagrangian motion in video analysis

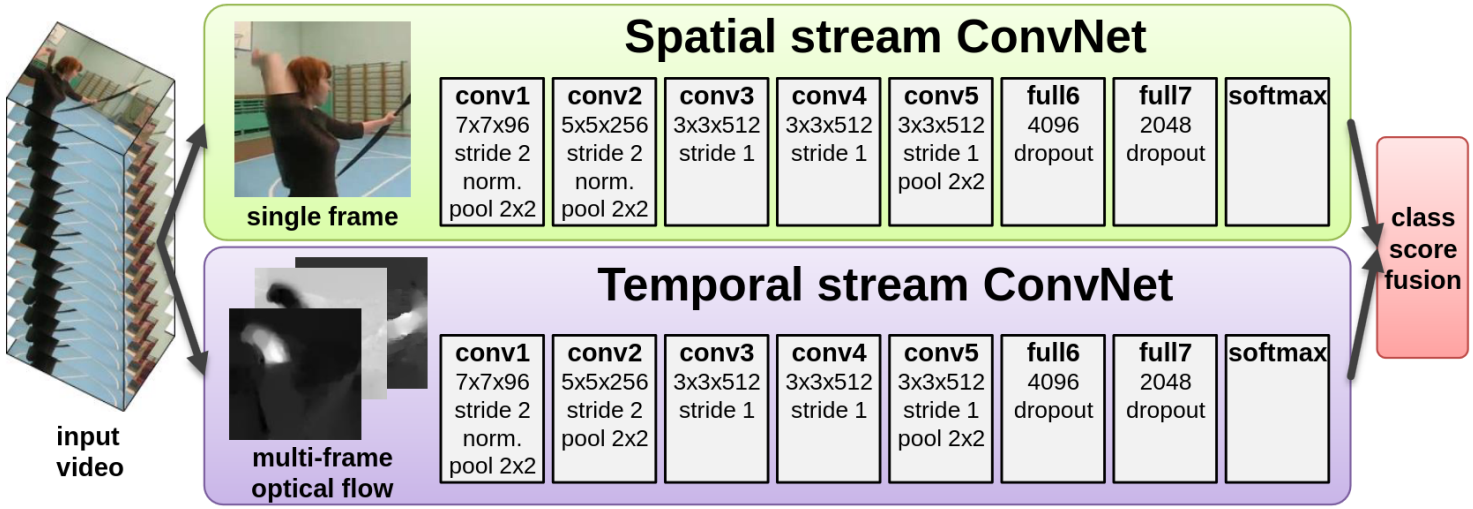
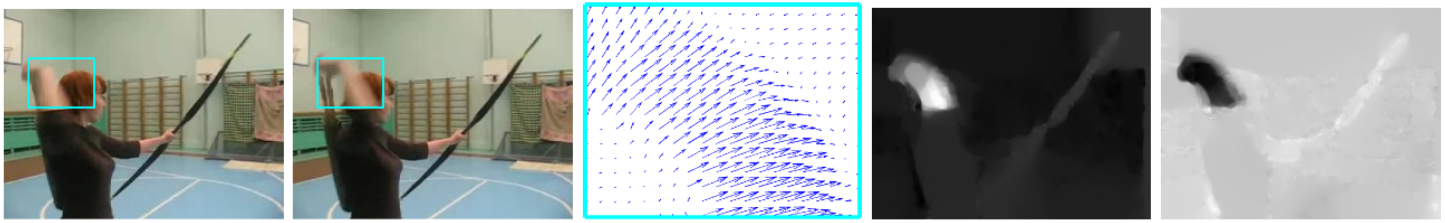


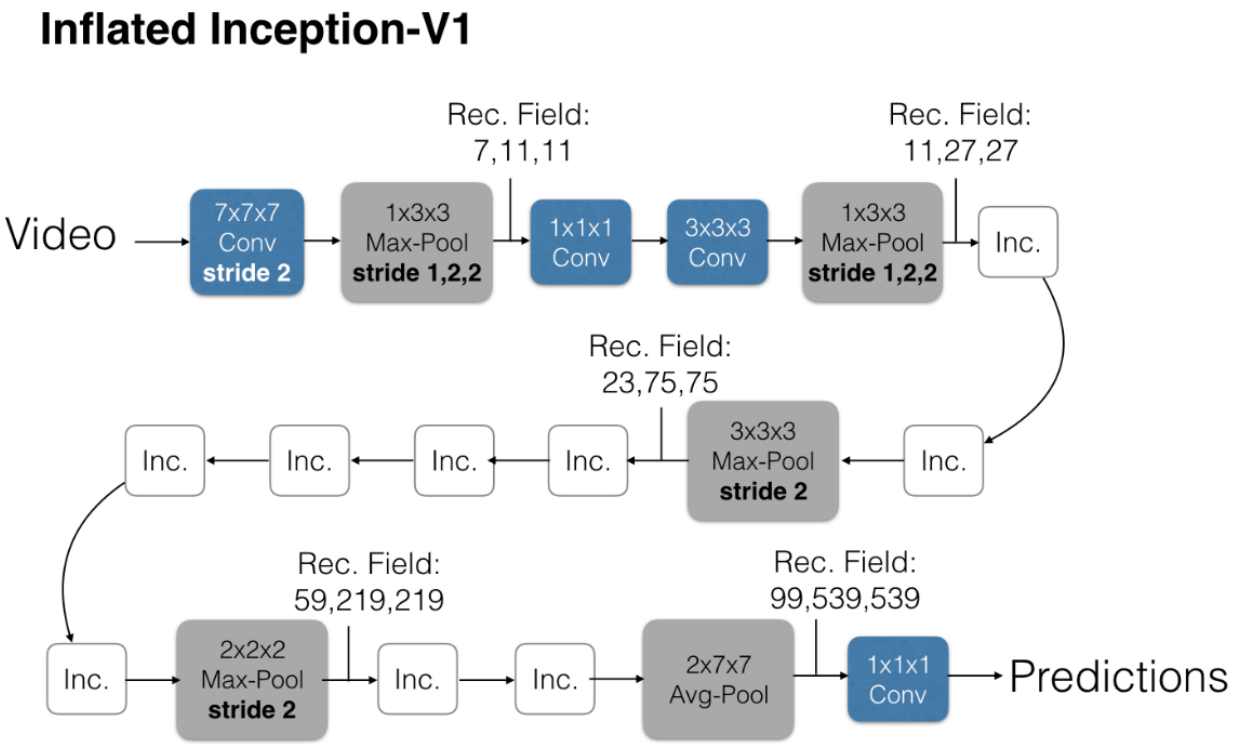
Figure 1: **Two-stream architecture for video classification.**

[K Simonyan, and A Zisserman. *"Two-stream convolutional networks for action recognition in videos."* NeurIPS, (2014)]



The best of both worlds

- Two-stream (OF and RGB) on *"inflated"* Inception backbone
- *Inflated 3D convolutions*: start from an ImageNet pre-trained Inception model and "inflate" the kernels on the time dimension



[J Carreira, A Zisserman. *"Quo vadis, action recognition? a new model and the kinetics dataset."*. CVPR 2017]

Follow-up methods

- Focus on more efficient methods for video analysis.

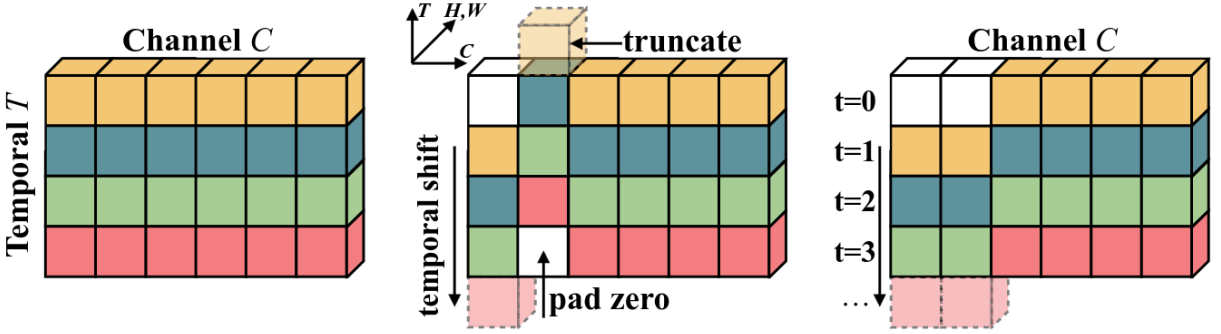
Why is video analysis inefficient?

Follow-up methods

- $3D$ convolutions are expensive, how can we approximate them?

Follow-up methods

- 3*D* convolutions are expensive, how can we approximate them?
- 2*D* convolutions + temporal shifts



(a) The original tensor without shift.

(b) Offline temporal shift (bi-direction).

(c) Online temporal shift (uni-direction).

Lin, Ji, Chuang Gan, and Song Han. *"Tsm: Temporal shift module for efficient video understanding."* ICCV (2019)]

Follow-up methods

- Videos have thousands of frames. How to efficiently process them?

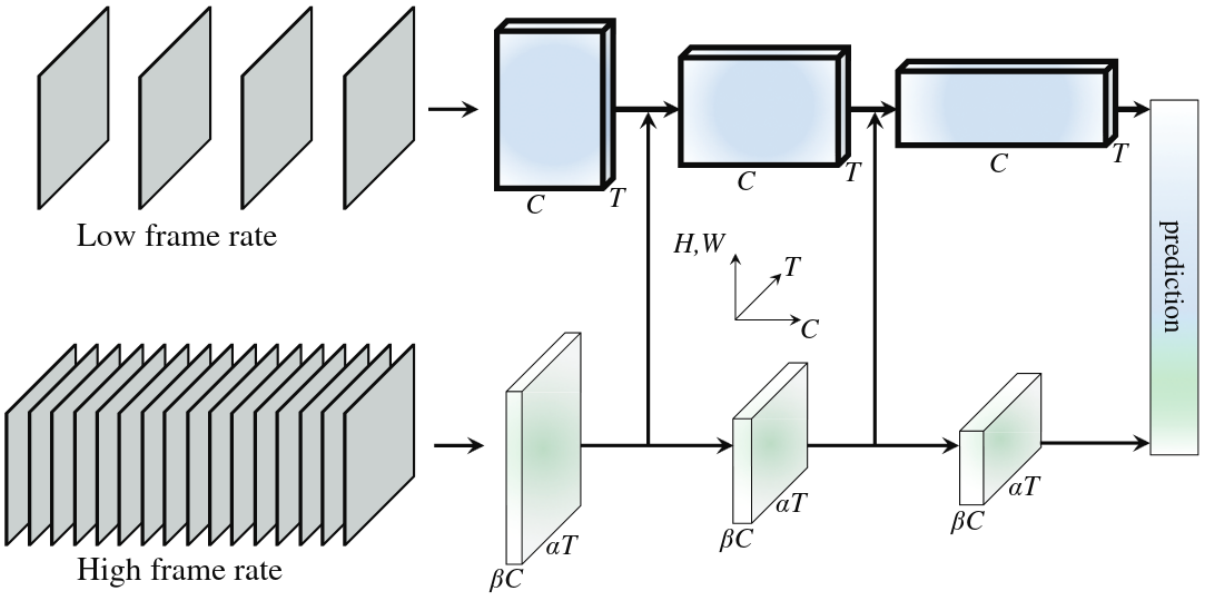


Figure 1. **A SlowFast network** has a low frame rate, low temporal resolution *Slow* pathway and a high frame rate, $\alpha\times$ higher temporal resolution *Fast* pathway. The Fast pathway is lightweight by using a fraction (β, *e.g.*, 1/8) of channels. Lateral connections fuse them.

Feichtenhofer, Christoph, et al. *"Slowfast networks for video recognition."* ICCV. (2019).

Follow-up methods

- Inflating 2*D* (ResNet archtectures) to 3*D* video models

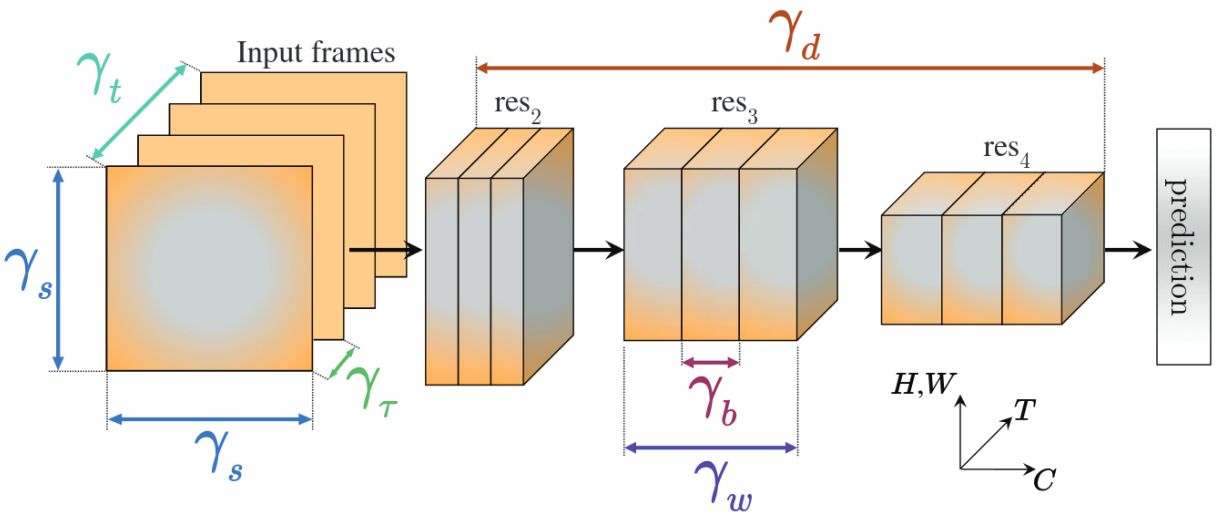
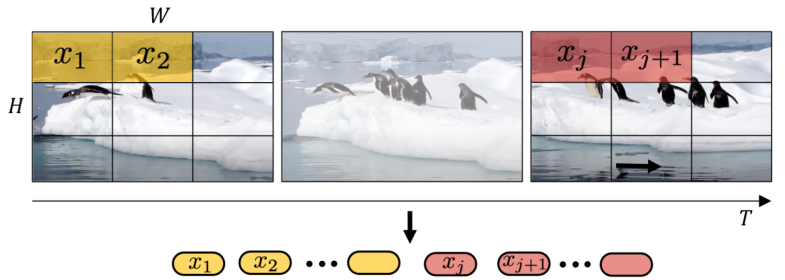


Figure 1. **X3D** networks progressively expand a 2D network across the following axes: Temporal duration γ_t, frame rate γ_τ, spatial resolution γ_s, width γ_w, bottleneck width γ_b, and depth γ_d.

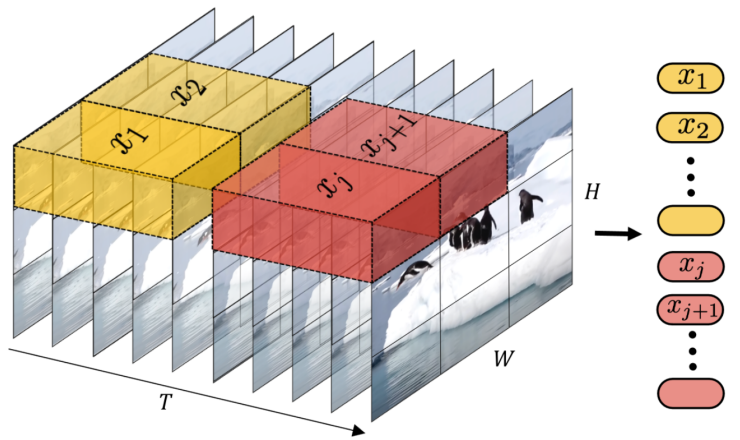
Feichtenhofer, Christoph. *"X3d: Expanding architectures for efficient video recognition."* CVPR (2020)

Different model flavors: Transformers

- *Factored transformer*: sample frame patches, and learn factored spatial and temporal encodings



- *Joint transformer*: sample video tubes, and learn joint spatio-temporal encodings



[Arnab, Anurag, et al. *"Vivit: A video vision transformer."* ICCV (2021)]

Different model flavors: Swin Transformers

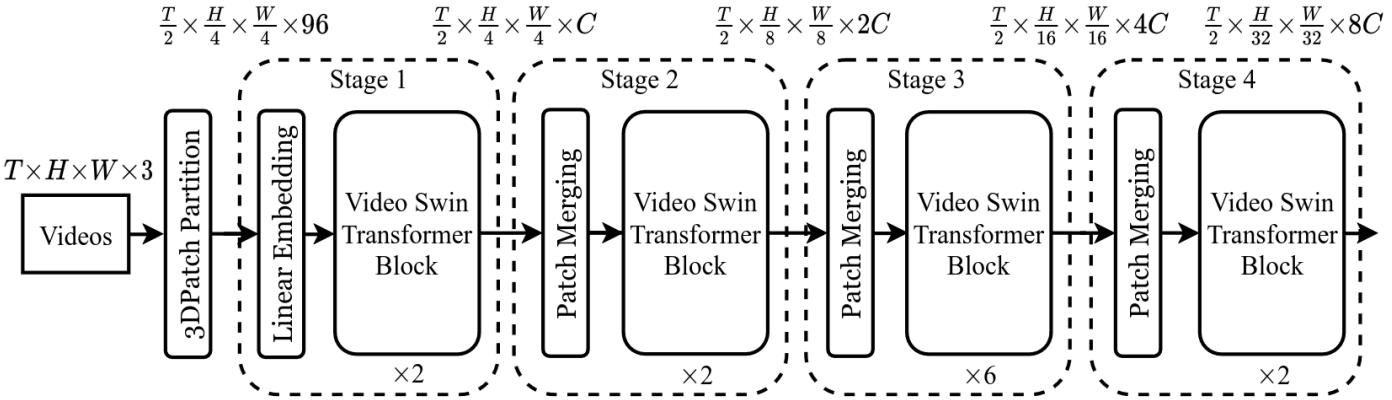
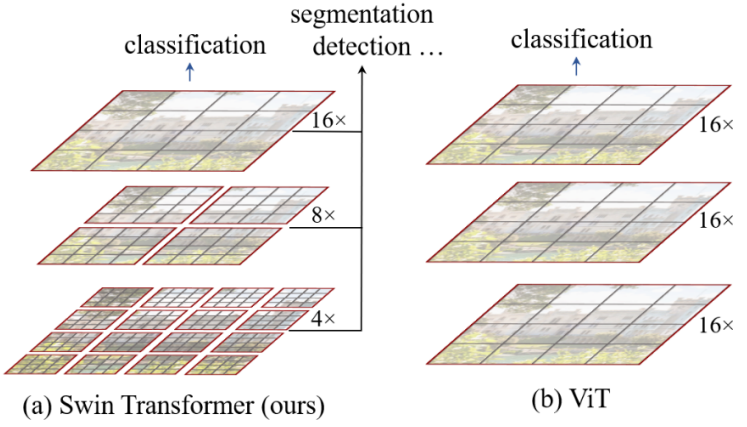
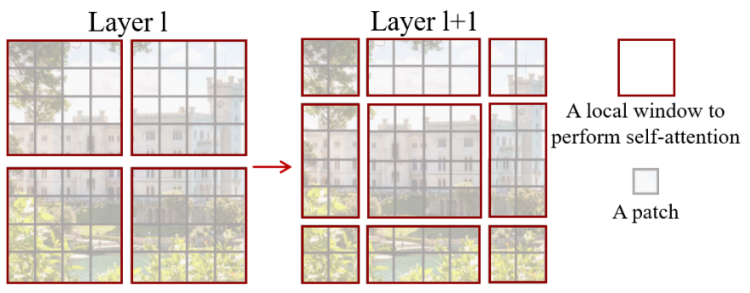


Figure 2. Overall architecture of Video Swin Transformer (tiny version, referred to as Swin-T).

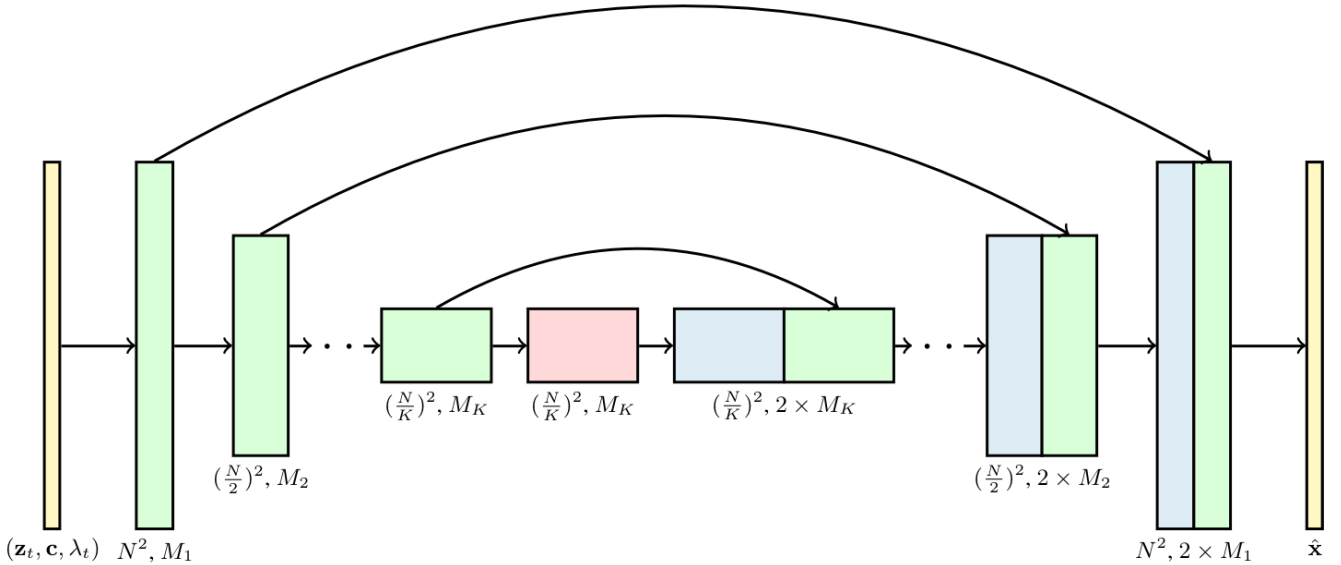




[Liu, Ze, et al. *"Video swin transformer."* CVPR. (2022)].

Different model flavors: Diffusion

- From the authors of the original DDPM



(Uniformative figure: Not plotted here, but they use spatial and temporal attention blocks)

[Ho, Jonathan, et al. *"Video diffusion models."* NeurIPS (2022).]

Questions? / Discussion